6.2.1. Evolution in theory

If you remember from your high school biology class, natural selection selects for the "most fit" individuals from each generation. In biology this represents the individuals that had the greatest reproductive success, and hence passed on their genetic information to subsequent generations. Birds with beak shapes more adept at procuring seeds from trees would have more food and thus be more likely to survive to pass that beak shape gene to their children and grandchildren. But remember, "most fit" is relative to an environment. A polar bear is well adapted to the polar ice caps but would be very unfit in the Amazonian rainforests. You can think of the environment as determining an objective or fitness function that assigns individuals a fitness score based on their performance within that environment; their performance is determined solely by their genetic information.

In biology, each mutation very subtly changes the characteristics of the organism, such that it may be difficult to discern one generation from another. However, allowing these mutations and variations to accumulate over multiple generations allows for perceptible changes. In the evolution of birds' beaks, for example, a population of birds would initially have had roughly the same beak shape. But as time progressed, random mutations were introduced into the population. Most of these mutations probably did not impact the birds at all or even had a deleterious effect, but with a large enough population and enough generations, random mutations occurred that affected beak shapes favorably. Birds with better suited beaks would have an advantage getting food over the other birds, and therefore they'd have a higher likelihood of passing down their genes. Therefore, the next generation would have an increased frequency of the favorably shaped beak gene.

In *evolutionary reinforcement learning*, we are selecting for traits that give our agents the highest reward in a given environment, and by *traits* we mean model parameters (e.g., the weights of a neural network) or entire model structures. An RL agent's fitness can be determined by the expected reward it would receive if it were to perform in the environment.

Let's say agent A played the Atari game Breakout and was able to achieve an average score of 500 while agent B was only able to obtain 300 points. We would say that agent A is more fit than agent B and that we want our optimal